BCL1213 Operating Systems

Mock Examination: Chapters 1 to 4

Practice paper | 11 November 2026

Time allowed: 2 hours
Total marks: 100
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Instructions

1. Answer all questions. Section A contains 20 questions worth 1 mark each. Section B contains four questions worth 20 marks each.
2. Choose one answer for each multiple choice question. There is no negative marking.
3. Show your working for all calculations. State units and label Gantt chart boundaries clearly.
4. Use a separate answer booklet. A basic calculator may be used. Attempt the paper without notes before checking the separate answer key.
5. Unless a question states otherwise, use only the assumptions given in that question. Do not infer priorities, time quanta or arrival times.

Coverage and suggested timing

Component	Marks	Suggested time
Section A: Questions 1 to 20	20	20 minutes
Question 21: Chapter 1	20	25 minutes
Question 22: Chapter 2	20	25 minutes
Question 23: Chapter 3	20	25 minutes
Question 24: Chapter 4	20	25 minutes
Total	**100**	**120 minutes**

Each chapter contributes 5 marks in Section A and 20 marks in Section B, for 25 marks per chapter.

Practice status: This is an independently prepared revision paper based on the course topics. It is not an official SEGi examination or a prediction of the actual examination format.

Section A: Multiple Choice

20 marks. Choose one answer for each question.

1. Which statement best describes an operating system?

A. A document editor that replaces all other programs
B. A network cable connecting two computers
C. A physical component used only for permanent storage
D. Software that manages hardware resources and provides services to applications

2. What is the main role of a device driver?

A. Store all running processes permanently
B. Allocate every CPU time slice
C. Replace the physical device
D. Translate OS requests into commands for a particular device

3. Which OS design places many core services in kernel space?

A. Microkernel
B. Time sharing
C. Monolithic kernel
D. Batch processing

4. Which statement distinguishes time sharing from basic multiprogramming?

A. Time sharing guarantees hard real time deadlines
B. Time sharing avoids all context switching
C. Time sharing emphasizes interactive response through frequent CPU allocation
D. Multiprogramming always requires several processors

5. Which application most clearly requires a hard real time response?

A. A safety controller that must react before a mandatory deadline
B. An overnight report with no strict finish time
C. A photo archive read occasionally
D. A document saved whenever the user chooses

Section A: Continued

6. A blocked process completes its awaited I/O. In the basic five state model, its next state is normally:

A. Terminated
B. New
C. Running immediately in every case
D. Ready

7. Which information is normally stored in a Process Control Block?

A. A physical CPU core
B. Every byte of every open file
C. A complete duplicate of the operating system
D. Saved program counter and CPU register values

8. What is a likely effect of an excessively small Round Robin quantum?

A. The scheduler becomes FCFS immediately
B. No process can be preempted
C. More frequent context switches and higher overhead
D. All CPU bursts become shorter

9. A process arrives at 2 ms, completes at 11 ms and uses 6 ms of CPU time. Assuming one CPU burst and no I/O, its waiting time is:

A. 3 ms
B. 11 ms
C. 9 ms
D. 6 ms

10. What property do threads of the same process normally share?

A. Exactly the same CPU register values
B. Their current program counter values
C. The process address space
D. Exactly the same stack

Section A: Continued

11. Which is the usual order from fastest to slowest access?

A. Registers, cache, RAM, SSD
B. Cache, SSD, registers, RAM
C. RAM, cache, registers, SSD
D. SSD, RAM, cache, registers

12. Unused space inside an allocated fixed size block is called:

A. External fragmentation
B. Internal fragmentation
C. Deadlock
D. Thrashing

13. In paging, a frame is:

A. The offset part of every instruction
B. A fixed size unit of physical memory
C. A variable size logical function
D. A permanently stored program file

14. A segment has base 2000 and length 500 bytes. Which offset is invalid?

A. 0
B. 500
C. 499
D. 125

15. What is thrashing?

A. A process finishing early
B. Excessive paging that leaves little time for useful execution
C. A kernel changing a file name
D. All memory becoming permanent storage

Section A: Continued

16. A race condition occurs when:

A. A variable is never modified
B. The result of shared state access depends on execution timing or order
C. Only one process exists
D. Every access is properly serialized

17. Which ownership rule normally applies to a mutex?

A. It must be released by an unrelated process
B. Ownership never exists
C. Any thread may always release another thread’s mutex
D. The acquiring thread must release it

18. A counting semaphore represents two available devices. Two threads acquire one permit each. A third acquire before either release will:

A. Wait for a permit to become available
B. Force both current users to terminate
C. Succeed because the initial value was positive
D. Create a new device

19. Which practice is correct when using a condition variable?

A. Recheck the shared predicate in a loop after waking
B. Treat each notification as a permanently stored permit
C. Access protected state without any mutex
D. Hold the mutex forever while sleeping

20. Requiring all threads to acquire locks in the same global order directly prevents:

A. CPU scheduling
B. Mutual exclusion
C. Circular wait
D. The existence of critical sections

Section B: Structured Questions

Question 21: OS Fundamentals and Architecture [20 marks]

(a) Define an operating system and explain two resource management responsibilities. **[5 marks]**

(b) A student application prints a document. Explain how the application, OS, device driver and printer cooperate to complete the request. **[4 marks]**

(c) Compare monolithic and microkernel designs in terms of where services execute and one trade off involving performance or fault isolation. **[6 marks]**

(d) A hospital system handles overnight billing and controls a device with a mandatory response deadline. Recommend an appropriate processing model for each task and justify both recommendations. Explain why a fast average response time does not prove that a hard deadline will be met. **[5 marks]**

Write your answers in the answer booklet.

Question 22: Process Management and Scheduling [20 marks]

A single CPU executes the following processes. Each process has one CPU burst and performs no I/O. Ignore context switching cost. If arrivals tie, use process number order.

Process	Arrival time (ms)	CPU burst (ms)
P1	0	6
P2	1	3
P3	2	1

(a) Using First Come, First Served (FCFS), draw the Gantt chart and calculate the completion time, turnaround time and waiting time for every process. Calculate the average waiting time. **[8 marks]**

(b) Repeat the Gantt chart and calculate each process completion time and waiting time using Round Robin with a quantum of **2 ms**. Calculate the average waiting time. If an arrival coincides with quantum expiry, enqueue that arrival before requeuing the process whose quantum expired. **[8 marks]**

(c) Distinguish Ready from Waiting in the process state model. Explain why a PCB is needed during a context switch. **[4 marks]**

Show all calculations and label the time boundaries.

Question 23: Memory Management [20 marks]

(a) A byte addressed system uses paging with a page size of **1024 bytes**. A process references logical address **2500**. Its page table includes page 0 mapped to frame 4, page 1 mapped to frame 2, and page 2 mapped to frame 7. Determine the page number, offset and physical address. Show how you validate the offset. **[6 marks]**

(b) A segmented system uses the table below. The limit denotes the segment length in bytes. For each reference, state whether it is valid. Calculate a physical address only for valid references and explain any invalid reference. **[6 marks]**

Segment	Base address	Limit (bytes)
0	1000	400
1	2500	600
2	4000	300

References: (i) segment 0, offset 200; (ii) segment 1, offset 600; (iii) segment 2, offset 299.

(c) Free holes in increasing address order are **100, 500, 200, 300 and 600 KiB**. A process requests **212 KiB**. Evaluate First Fit, Best Fit and Worst Fit independently using the original hole list. For each, state the chosen hole and remaining size. **[6 marks]**

(d) Distinguish internal fragmentation from external fragmentation. **[2 marks]**

Question 24: Synchronization and Deadlocks [20 marks]

(a) A shared integer `counter` initially equals 10. Thread A performs `counter = counter + 1`, and Thread B performs `counter = counter − 1`. The operations are not atomic. The following interleaving occurs:

```
1. A reads counter into its local register.
2. B reads counter into its local register.
3. A adds 1 to its register.
4. B subtracts 1 from its register.
5. A writes its register to counter.
6. B writes its register to counter.
```

State the final counter value, the result expected if both updates were serialized, and the concurrency problem. Describe how a mutex should protect the updates. **[6 marks]**

(b) A connection pool has three identical connections. Explain how a counting semaphore initialized to 3 controls access, including what happens when a fourth thread requests a connection before any of the first three releases one. Distinguish this semaphore from a mutex in terms of ownership. **[4 marks]**

(c) A consumer waits until a shared queue is nonempty. Explain how a condition variable and mutex should be used. Your answer must describe mutex release, reacquisition and the need for a predicate loop. **[4 marks]**

(d) Thread X holds lock A and requests lock B. Thread Y holds lock B and requests lock A. Name the four necessary conditions for resource deadlock. Explain how a consistent global acquisition order prevents this situation. **[6 marks]**

End of paper